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Lesson plan

Name of the Programme:	**B.Tech(Hons.)**	
Title of the Course:	**Embedded System and ARM Microcontroller**	
Course code	**CS1120**	
Semester:	**II**	
Names of the Course Instructor(s)	**Dr. Vidya M J, Dr. Sheela S, Prof.Shilpa A. Hiremath, Prof. Anushree M K, Dr. Sumathi H R, Dr. Nagaraj J, Dr. Shouvik Chakraborthy**	
Course credit	**No. of hours per week (2+0+2)**	**Total no. of Teaching hours**
03	**04 Hours**	**60 Hours**

Session No.	Unit	Topic / Lab Detail	Pedagogy / Activities	Reference / Pre-reading
1	Unit1	Introduction to Embedded Systems	Lecture	ARM SoC Design - Ch.1
2		Design Challenges of Embedded Systems	Discussion	ARM SoC Design - Ch.1
3		Lab 1a: Blink LED_BUILTIN	Hands-on	Lab Manual Exp 1a
4		Lab 1a (cont.): Serial Print 'Hello World'	Hands-on	Lab Manual Exp 1a
5		Types of Memory	Lecture	RP2040 Datasheet
6		Memory Mapping of a C Program	Diagram + IDE mapping	Embedded C Manual
7		Lab 1b: Alternate LED Blink (3s/5s delay)	Coding & Simulation	Lab Manual Exp 1b
8		Lab 2a: LED Binary Count 1–15	Loop control	Lab Manual Exp 2a
9		Stack, Code, Data Overview	Lecture + diagrams	Embedded C Memory Model

10		Lab 2a (cont.): Count back 15–1	Binary pattern simulation	Lab Manual Exp 2a
11		GPIO Concepts	Block diagram explanation	RP2040 Manual
12		Lab 2b: Odd/Even LED Display	Bitwise logic	Lab Manual Exp 2b
13		Case Study: Embedded Memory Systems	Group discussion	Case Study Sheet
14		Lab Review and Q&A	Recap of all Unit 1 labs	Lab Notebook
15	Unit2	ARM Architecture: Cortex A/R/M	Lecture	ARM Architecture Guide
16		Cortex-M Programmer Model	Registers overview	Cortex-M TRM
17		Lab 3a: Return 200 + Sum of 3 Constants	Inline ASM	Lab Manual Exp 3a
18		Lab 3b: Sum of 4 Constants	Hands-on	Lab Manual Exp 3b
19		Lab 3c: Sum of 5 Constants	Hands-on	Lab Manual Exp 3c
20		AAPCS Overview	Calling convention diagram	AAPCS Manual
21		Lab 4a: ADD 3 Constants in ASM	Coding	Lab Manual Exp 4a
22		Lab 4b: AND/OR Logic	Instruction testing	Lab Manual Exp 4b
23		Lab 4c: NOT Logic	Hands-on	Lab Manual Exp 4c
24		CMP & Branching	Instruction walkthrough	Branching Examples
25		Lab Review	Debug and discussion	Lab Review Sheet

26		GPIO + ASM Integration Concepts	Review & Case Study	Integration Notes
27		Mini Quiz / Oral Viva	Quick assessment	Unit 2 Questions
28		Lab Feedback & Logs	Peer review	Lab Reflection Sheet
29	Unit3	Shift Instructions: LSR, LSL	Code demo	ARM Instruction Manual
30		Lab 5a: 2x LSR Operation	Code execution	Lab Manual Exp 5a
31		Lab 5b: 4x LSL Operation	Verification	Lab Manual Exp 5b
32		Arithmetic Shift Operation (ASR)	Signed example	ASR Theory
33		Lab 6: 8x ASR on Negative Number	Bit simulation	Lab Manual Exp 6
34		CMP, Flags, PSR Explanation	Demo + trace	Flag Register Docs
35		Lab 7a: DeMorgan Verification	Coding logic	Lab Manual Exp 7a
36		Lab 7b: XOR Truth Table Verification	Hands-on	Lab Manual Exp 7b
37		Branching Practice with EQ/NE	Flow control	Flowchart Sheet
38		Function Call Examples	Inline stack trace	Function Debugging
39		Nested Looping in Assembly	Discussion	Looping Sheet
40		Lab Debugging: PSR + Flags	Debugging session	Lab Worksheet
41		PSR Use in Real Programs	Viva discussion	PSR Usage

42		Application Task using CMP	Logic design	CMP Examples
43		Lab Quiz	Assessment	Quiz Sheet
44		Oral Viva	Instructor-led viva	Viva Questions
45	Unit4	C-ASM Integration,),Developing programs using CMP, LDR, STR, and logical instructions.	Parameter walkthrough	Lab codes execution
46		Lab 8a: ADD Two Unsigned Integers	Function implementation	Lab Manual Exp 8a
47		Lab 8b: Read PSR + Return to C++	Hands-on coding	Lab Manual Exp 8b
48		CMP Logic – EQ, LO, HI, GE	Conditions diagram	CMP Sheet
49		Lab 9a: CMP EQ – Logic Test	Code validation	Lab Manual Exp 9a
50		Lab 9b: CMP LO – Branching	Simulate and test	Lab Manual Exp 9b
51		Application Binary Interface (ABI) and AAPCS (ARM Procedure Call Standard	Datasheet explanation	Lab Manual with example codes explanation
52		Lab 10b: CMP GE – Signed Comparison	Final hands-on	Lab Manual Exp 10b
53		Final Recap – Logic Flow	Board activity	Review Sheet
54		Lab Integration Summary	Reflect + write up	Notebook
55		LDR,STR Explanation	Code assessment	Lab codes
56		Debug & Optimize	Efficiency testing	Debug Sheet

57	Unit5	Push pop instruction with assembly instructions	Coding examples	Wowki simulation
58		I2C,DMA explanation	With sensors integration explanation of protocols	Datasheet reference to understand how to interface with sensors
59		ISR explanation	With UART example	Demo of UART code as ISR routine
60		Revision and Course Exit & Feedback	Survey + feedback	Exit Survey Form

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component	20 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab Continuous Evaluation - Project Work - Lab Test	5 marks out of 70 10 marks out of 70 10 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory

Course Outcomes to Program Outcome Mapping

Program Outcome / Course Outcome		**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**
CO1	Understand the concept of embedded system,	3	2	-	-	1	-	-	-	2	-	-	3

	microcontroller, different components of microcontroller and their interactions as applied to RP2040.												
C02	Get familiarized with programming environment to develop embedded solutions.	3	2	2	1	3	-	-	-	1	-	-	3
C03	Program ARM microcontroller to perform various tasks and stack implementation in Cortex-M0+.	3	2	3	1	3	-	-	-	2	2	-	3
CO4	Understand the key concepts of embedded systems such as Instruction cycle execution, I/O, DMA, interrupts and interaction with peripheral devices.	3	2	-	1	2	-	-	-	-	-	-	3

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Quiz Graded Component 1	Quiz will be conducted for 10 marks. Each right answer carries one mark each. No negative marking for wrong answers.
	Surprise Test	A Surprise Test will be conducted for 10 marks which shall include MCQs, True (or) False, One word answer or

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	Graded Component 2	writing a simple code

Total Marks		25
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		25
CIE-3 Components		**Rubrics for Assessment**
CIE-3	Continuous Weekly Lab Conduction	Lab Conduction = 5 marks ***(Refer Table-2)***
	Project Work	Project Exhibition = 10 marks ***(Refer Table-3)***
	Lab Test	Lab Test =10 marks ***(Refer Table-4)***

Table2: **Rubrics for Lab Write-up and Execution (Max: 5+5 = 10 marks)**						
LAB WRITE-UP (5 Marks)						
Sl.No.	**Criteria**	**Measuring Method**	**Excellent**	**Good**	**Satisfactory**	**Poor**
1	Lab Record write up **(1 Marks)**	Submission of lab record	Complete write up of Lab programs with proper syntax. **(1 Marks)**	Complete write up of Lab programs with minor issues **(0.75 Mark)**	Complete write up of Lab programs with some syntax errors**(0.5 Marks)**	No submission of Lab Record **(0 Marks)**
2	Interfacing Hardware	Observation	The hardware circuit is	The circuit functions correctly	The circuit is partially functional	The circuit is incorrect,

	board **(1 Marks)**		assembled accurately, neatly, and operates efficiently as intended.**(1 Mark)**	with minor wiring or implementation issues.. Results are interpreted correctly. **(0.75 Mark)**	and shows noticeable errors or inefficiencies. **(0.5 Marks)**	unsafe, or non-functional. Results are misinterpreted. **(0 Marks)**
3	Software Developme nt (2 Mark)	Code Review	The program code is efficient, well-structured, fully functional, and clearly commented. **(2 Mark)**	The code functions correctly with minor inefficiencies or limited documentation. **(1.5Marks)**	The code is functional but lacks efficiency, clarity, or sufficient commenting. **(1 Mark)**	The code is inefficient, poorly documented, or non-functional.. **(0 Mark)**
4	Interpretatio n of Result and Insight **(1 Mark)**	Observation	Results are interpreted insightfully, clearly linked to the experiment objectives, and demonstrate strong conceptual understanding.. **(1 Mark)**	Interpretation is clear and mostly aligned with the objectives. **(0.75 Marks)**	Interpretation is adequate but shows limited connection to the objectives.. **(0.5 Mark)**	Interpretation is unclear and poorly related to the objectives. **(0 Mark)**

VIVA VOCE (5 Marks)						
Sl. No.	**Criteria**	**Measuring Method**	**Excellent**	**Good**	**Satisfactory**	**Poor**
1	Understan ding of Concepts **(3 Marks)**	Viva	Demonstrates a thorough understanding of core concepts and explains them clearly and confidently. **(3 Marks)**	Shows good understanding of core concepts with minor misconceptions. **(2 Marks)**	Demonstrates partial understanding of core concepts with several misconceptions. **(1 Mark)**	Lacks understanding of core concepts and is unable to explain them clearly. **(0 Mark)**

2	Response to Questions **(2 Marks)**	Viva	Responds accurately, confidently, and thoughtfully, providing detailed and well-reasoned answers.. **(2 Marks)**	Responds accurately and confidently, with minor details missing. **(1.5 Mark)**	Responds with some inaccuracies or hesitation; answers are often incomplete or lack sufficient detail.**(1 mark)**	Struggles to respond accurately or confidently; answers are largely incorrect, incomplete, or unclear **(0 Mark)**

Note: Evaluation is based on students' lab performance every week . Each experiment is evaluated for 10 marks which sums upto 100 marks by the end of semester and later scaled down to 5 marks

Table 3 : Project Exhibition Rubric (10 Marks)

Sl.No.	**Component**	**Maximum Marks**	**Description**
1	Concept & Understanding	3 Marks	3 – Concept is very clear; student explains the project confidently and accurately 2 – Concept is mostly clear with minor gaps 1 – Limited understanding of the project
2	Creativity & Innovation	2 Marks	2 – Project shows originality and creative thinking 1 – Some creativity shown
3	Presentation & Model/Display	3 Marks	3 – Neat, attractive, well-organized display/model 2 – Good presentation with small issues 1 – Poor or unclear presentation
4	Communication & Interaction(Q &A)	2 Marks	2 – Explains clearly, answers questions confidently 1 – Explanation is partially clear

Note: Evaluation is done for Individual Student

Table 4: Lab Test Rubrics (10 Marks)

Sl.No	**Component**	**Criteria**	**Marks**	**Description**
1.	Write-up with Correct Code	Clarity of explanation, proper structure, correct syntax, and relevance to lab objectives	5 marks	- 5: Accurate code, well-documented with clear logic - 4: Minor syntax/logic errors, mostly complete write-up - 3-2: Code lacks clarity or has several mistakes - 1–0: Incomplete or mostly incorrect code/no write up.
2.	Code Execution and Hardware Interfacing	Hardware / software Functional output without errors	10 marks	10-9: Fully functional code on Hardware with correct output . - 8: Minor issues or missing features, but core task works - 7-5: Partial Execution. - 4-2: Code runs with major issues or incorrect output

				- 1-0: Code doesn't compile or execute due to errors
3.	Viva (Oral Questions)	Conceptual understanding, ability to justify logic , responsiveness	5 marks	5: Answers confidently, shows clear understanding - 4: Understands basics but struggles with application - 3-1: Struggles with basic concepts - 0: Unable to answer clearly or justify code decisions

Note: Evaluation is done for 20 marks which would be scaled down to 10 marks

Signature of Faculty